

WHAT ARE ANIMALS?

With almost 2 million species identified to date, and even more than that awaiting discovery, animals are the most varied living things on the planet. For over a billion years, they have adapted to the changing world around them, developing a vast array of different lifestyles in the struggle to survive. At one extreme, animals include fast-moving predators, such as sharks, big cats, and birds of prey; at the other, there are the inconspicuous sorters and sifters of the animal world's leftovers, living unseen in the soil or on the deep seabed. Together, they make up the animal kingdom—a vast collection of living things that are linked by a shared biology and that occupy a dominant place in life on Earth.

Characteristics of animals

Most animals can move faster than other kinds of organisms. Although not all animals move from place to place, or locomote, the combination of muscles and nerves of animals means that they can respond quickly to the world around them and move at least parts of their bodies by contracting their muscle fibers. Most important of all, they get the energy they need by taking in food.

Animals are highly complex and remarkably responsive, compared with other forms of life. Even the simplest animals react quickly to changes around them, shrinking away from potential danger or reaching out for food. Animals with well-developed nervous systems can go much further: they can learn from experience—an ability that is unique to the animal world.

The scale of animal life

The world's largest living animals, baleen whales, can be up to 82 ft (25 m) long and weigh 132 tons (120 tonnes). At the other end of the

spectrum are microscopic organisms—rotifers and tardigrades are only $\frac{1}{500}$ in (0.05 mm) long—and submicroscopic flies and beetles about $\frac{1}{125}$ in (0.2 mm) long. These animals are so tiny that their weight is negligible. Even so, they possess all the body systems needed for survival.

Different body sizes allow animals to live in different ways. Whales have few natural predators; and the same is true of elephants, the largest land animals. Their massive bodies are highly energy-efficient because they process food on such a large scale. However, they take a long time to reach maturity, which means they are slow to reproduce. Insects, on the other hand, are easy prey for many animals, and their small size means that their bodies are not as energy-efficient as those of large animals. But, since they can breed rapidly when conditions are favorable, their numbers can climb at a prodigious rate.

Chordates and invertebrates

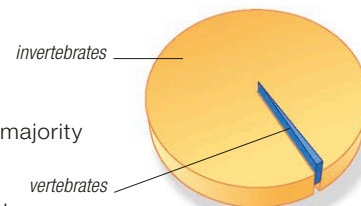
Almost all the world's largest and most familiar animals are vertebrates—animals with internal skeletons made of cartilage or bone. They include the fastest animals on land, in water, and in the air, and also many species with elaborate behavior and highly developed brains. Vertebrates themselves belong to a group of animals called chordates, which share a common ancestry stretching back millions of years. However, despite leading the animal kingdom in many fields, chordates make up a tiny minority of the animal species

alive today. The vast majority of species are invertebrates, or animals without backbones.

Unlike chordates, invertebrates often have very little in common with each other, apart from their lack of a backbone. The giant squid, the largest invertebrate, can measure over 52 ft (16 m) long, but it is very much an exception. Most invertebrates

INVERTEBRATE MAJORITY

Chordates make up less than 3 percent of the world's animals. The remaining 97 percent are invertebrates.

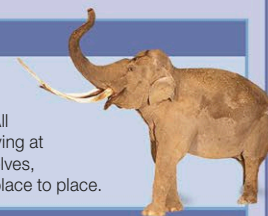


KINGDOMS OF LIFE

Biologists classify all living things into overall groups, called kingdoms. The members of each kingdom are alike in fundamental ways, such as in the nature of their cells or in the way they obtain energy. In the most widely used system of classification there were 6 kingdoms, of which the animal kingdom was the largest. However, changes to this classification have split the prototists into multiple kingdoms, reflecting a deep-rooted diversity that encompasses plantlike, funguslike, and animallike forms, many of which are single-celled. An altogether different classification has also been proposed, which groups all life into 3 "superkingdoms": Archaeobacteria, Eubacteria, and Eukaryota.

ANIMALS

Animals are multicellular organisms that obtain energy by ingesting food. All animals are capable of moving at least some parts of themselves, and many can move from place to place.



PLANTS

Plants are multicellular organisms that grow by harnessing the energy in light. Through a process called photosynthesis, they use this energy to build up organic matter from simple materials, creating most of the food on which animals rely.



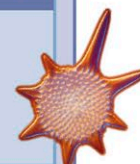
FUNGI

Most fungi are multicellular. They collect energy from organic matter, which they do not ingest but break down externally using microscopic threads that spread throughout their food. Many fungi are too small to be seen, but some form large fruiting bodies.



PROTISTS

Protists are single-celled organisms that typically live in water, or in permanently moist habitats. Their cells are larger and more complex than those of bacteria. Some protists behave like plants, collecting energy from sunlight; others, known as protozoa, are more like animals, acquiring energy by ingesting food.



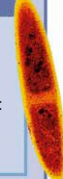
BACTERIA

Together with Archaea, bacteria are the simplest fully independent living things. Their cells are prokaryotic, meaning they lack organelles—the specialized structures that more complex cells use for carrying out different tasks. Bacteria gather energy from various sources, including organic and inorganic matter, and sunlight.



ARCHAEA

Single-celled archaeans used to be classified among bacteria, and like them they lack cellular organelles. But details of their chemical makeup link them more closely to higher organisms. Many archaeans are extremophiles: they are adapted to live in extreme environments, such as hot volcanic springs and super-saline lakes.



STATIC LIFE
Sea squirts are typical "sessile" animals: they spend their adult lives fixed to a solid surface. The young, which resemble tadpoles, can move freely, enabling sea squirts to spread.



Microscopic tardigrades (also called water bears) move about on tiny legs. Their behavior is as complex as that of animals thousands of times their size.



OCEAN GIANTS

A humpback whale bursts out of the sea. Animals this large depend on aquatic habitats because water can support most of their weight.